CS6450

AFL: A Single-Round Analytic Approach for Federated Learning with Pre-trained Models

(CVPR-2025)

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Summary of Paper

Pain Points of Conventional FL:

- Data Heterogeneity (non-IID) → divergence
- Large client numbers → performance collapse
- Slow convergence → hundreds of rounds
- High communication cost → bandwidth bottleneck

- **Problem Statement**

- The research objective is to develop a novel FL framework that overcomes these limitations by providing analytical, closed-form solutions for efficient and robust training

Summary of Paper

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Analytic Federated Learning

- We propose **Analytic Federated Learning (AFL)**, a gradient-free FL framework utilizing analytical closed-form solutions for both local client training and aggregation.

- AFL incorporates a **one-epoch client** training stage where a **pre-trained network's** classification head is reformulated into a localized linear regression problem, yielding an LS-based solution.

- The aggregation stage introduces an **Absolute Aggregation (AA) law,** derived analytically, enabling single-round aggregation without approximation or parameter tuning

Summary of Paper

1. Absolute Aggregation Law (AAL)

$$\widehat{W} = W_u \widehat{W}_u + W_v \widehat{W}_v$$

•Global model = weighted combination of local models
•Weights depend on:
- $C_k = X_k^T X_k$ data covariance)

•Exact centralized solution (no raw data sharing)

2. Multi-Client Extension

$$\widehat{W}_{agg,k} = W_{agg} \widehat{W}_{agg,k-1} + W_k \widehat{W}_k$$

•Aggregation is pairwise & iterative, not averaging

Limitation
•Requires full column rank
•Breaks when data is limited / rank-deficient

3. RI-AA Law (Fix with Regularization)

$$\widehat{W}_{agg,k} = \left(C^r_{agg,k} + k\gamma I\right)^{-1} C^r_{agg,k} \widehat{W}^r_{agg,k}$$

•Adds regularization to handle rank deficiency

Summary of Paper

Local Stage

Each client:

Compute:

$$\mathbf{X}_k, \mathbf{Y}_k$$

Solve:

$$\hat{\mathbf{W}}_k^r = (\mathbf{X}_k^T \mathbf{X}_k + \gamma \mathbf{I})^{-1} \mathbf{X}_k^T \mathbf{Y}_k$$

Compute:

$$\mathbf{C}_k^r = \mathbf{X}_k^T \mathbf{X}_k + \gamma \mathbf{I}$$

Server Stage

Step 1: Initialize

$$\hat{\mathbf{W}}_{\text{agg},0}^r = \mathbf{0}, \mathbf{C}_{\text{agg},0}^r = \mathbf{0}$$

Step 2: Aggregate

For each client:

$$\hat{\mathbf{W}}_{\text{agg},k}^r = \mathbf{W}_{\text{agg}} \hat{\mathbf{W}}_{\text{agg}-1} + \mathbf{W} \hat{\mathbf{K}}_k^r$$

$$\mathbf{C}_{\text{agg},K}^r = \mathbf{C}_{\text{agg},k-1}^r + \mathbf{C}_k^r$$

Step 3: Restore

$$\hat{\mathbf{W}} = (\mathbf{C}_{\text{agg},K})^{-1} \mathbf{C}_{\text{agg}} \hat{\mathbf{W}}_{\text{agg},K}$$

Implementation Details

- **Datasets:** CIFAR-10, CIFAR-100, Tiny-ImageNet

- **Backbone:** ResNet-18 pretrained on ImageNet-1k (frozen)

- **Data Partitioning:**
- NIID-1: Latent Dirichlet Allocation (LDA) — $\alpha = 0.1, 0.01$
- NIID-2: Sharding — $s = 10, 5$ (CIFAR-100/Tiny), $s = 4, 2$ (CIFAR-10)

- **Setup:**
- 100 clients per experiment (also tested 500, 1000)
- Batch size: 512

Performance Metrics Used

Primary Metric:
Top-1 Classification Accuracy (%) on held-out test set

Secondary Metrics:
- **Training Time (seconds)** — wall-clock time for full pipeline
- **Communication Rounds** — number of server-client aggregation cycles

Analysis Dimensions:
Accuracy vs. α (heterogeneity level) — data heterogeneity invariance
Accuracy vs. K (client count) — client-number invariance

Implementation - Results

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Invariance to Heterogeneity with Alpha = 0.1 and Alpha = 0.05

CIFAR - 10

Alpha = 0.1

```
Training Accuracy at training set on Client #99: 99.99999237060547%
Elapsing time for local training: 237.0807478427887%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 258.8226857185364%
Average accuracy on all Client: 82.5%
Average accuracy after regularization cleaning on all Client: 82.48999786376953%
Elapsing time plus cleansing regularization: 278.4478986263275%
written it to a csv file named cifar10resnet18_100_0.1.csv.
```

Alpha = 0.05

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 244.53295707702637%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 267.31181931495667%
Average accuracy on all Client: 82.5%
Average accuracy after regularization cleaning on all Client: 82.48999786376953%
Elapsing time plus cleansing regularization: 286.1822757720947%
written it to a csv file named cifar10resnet18_100_0.05.csv.
```

CIFAR - 100

Alpha = 0.1

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 263.19539070129395%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 284.6139307022095%
Average accuracy on all Client: 58.5099983215332%
Average accuracy after regularization cleaning on all Client: 58.55999755859375%
Elapsing time plus cleansing regularization: 303.7576811313629%
written it to a csv file named cifar100resnet18_100_0.1.csv.
```

Alpha = 0.05

```
Training Accuracy at training set on Client #99: 99.99999237060547%
Elapsing time for local training: 250.54799056053162%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 272.4004168510437%
Average accuracy on all Client: 58.5099983215332%
Average accuracy after regularization cleaning on all Client: 58.55999755859375%
Elapsing time plus cleansing regularization: 291.2432882785797%
written it to a csv file named cifar100resnet18_100_0.05.csv.
```

Tinyimagenet

Alpha = 0.1

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 520.7191429138184%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 544.459431886673%
Average accuracy on all Client: 54.779998779296875%
Average accuracy after regularization cleaning on all Client: 54.80999755859375%
Elapsing time plus cleansing regularization: 564.5403981208801%
written it to a csv file named tinyimagenetresnet18_100_0.1.csv.
```

Alpha = 0.05

```
Training Accuracy at training set on Client #99: 99.71098327636719%
Elapsing time for local training: 515.3877384662628%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 539.0812811851501%
Average accuracy on all Client: 54.779998779296875%
Average accuracy after regularization cleaning on all Client: 54.80999755859375%
Elapsing time plus cleansing regularization: 560.0547194480896%
written it to a csv file named tinyimagenetresnet18_100_0.05.csv.
```

https://colab.research.google.com/drive/1Hiqi46-bOY8U43E3KfudXAPEUm-u8aGp?usp=sharing

Implementation - Results

Invariance to Heterogeneity with CIFAR-10 (S = 4 and S = 2) CIFAR-100 and Tinyimagenet (S = 10 and S = 5)

CIFAR - 10

S = 4

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 238.40551161766052%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 260.2075307369232%
Average accuracy on all Client: 82.5%
Average accuracy after regularization cleaning on all Client: 82.48999786376953%
Elapsing time plus cleansing regularization: 278.283061504364%
written it to a csv file named cifar10resnet18_100_0.1.csv.
```

S = 2

```
Training Accuracy at training set on Client #99: 99.91596984863281%
Elapsing time for local training: 234.9704463481903%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 255.78718256950378%
Average accuracy on all Client: 82.5%
Average accuracy after regularization cleaning on all Client: 82.48999786376953%
Elapsing time plus cleansing regularization: 274.6025056838989%
written it to a csv file named cifar10resnet18_100_0.1.csv.
```

CIFAR - 100

S = 10

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 277.7598855495453%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 299.43948769569397%
Average accuracy on all Client: 58.5099983215332%
Average accuracy after regularization cleaning on all Client: 58.55999755859375%
Elapsing time plus cleansing regularization: 317.6767156124115%
written it to a csv file named cifar100resnet18_100_0.1.csv.
```

S = 5

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 269.23117446899414%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 292.1030867099762%
Average accuracy on all Client: 58.5099983215332%
Average accuracy after regularization cleaning on all Client: 58.55999755859375%
Elapsing time plus cleansing regularization: 312.10649275779724%
written it to a csv file named cifar100resnet18_100_0.1.csv.
```

Tinyimagenet

S = 10

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 510.8326425552368%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 535.0601077079773%
Average accuracy on all Client: 54.779998779296875%
Average accuracy after regularization cleaning on all Client: 54.80999755859375%
Elapsing time plus cleansing regularization: 556.571320772171%
written it to a csv file named tinyimagenetresnet18_100_0.1.csv.
```

S = 5

```
Training Accuracy at training set on Client #99: 100.0%
Elapsing time for local training: 652.6979072093964%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 681.893501996994%
Average accuracy on all Client: 54.779998779296875%
Average accuracy after regularization cleaning on all Client: 54.80999755859375%
Elapsing time plus cleansing regularization: 703.9910371303558%
written it to a csv file named tinyimagenetresnet18_100_0.1.csv.
```

https://colab.research.google.com/drive/1Hiqi46-bOY8U43E3KfudXAPEUm-u8aGp?usp=sharing

Implementation - Results

Invariance to No. of Clients with K = 100, K = 500 and K = 1000

CIFAR - 100

K = 500

```
Elapsing time for local training: 797.8273799419403%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 841.1920909881592%
Average accuracy on all Client: 58.32999801635742%
Average accuracy after regularization cleaning on all Client: 58.55999755859375%
Elapsing time plus cleansing regularization: 859.4695720672607%
written it to a csv file named cifar100resnet18_500_0.1.csv.
```

K = 1000

```
Training Accuracy at training set on Client #999: 100.0%
Elapsing time for local training: 826.0462906360626%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 879.507586479187%
Average accuracy on all Client: 58.14999771118164%
Average accuracy after regularization cleaning on all Client: 58.55999755859375%
Elapsing time plus cleansing regularization: 898.1180558204651%
written it to a csv file named cifar100resnet18_1000_0.1.csv.
```

Tinyimagenet

K = 500

```
Elapsing time for local training: 1163.0233738422394%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 1206.4319450855255%
Average accuracy on all Client: 54.75%
Average accuracy after regularization cleaning on all Client: 54.80999755859375%
Elapsing time plus cleansing regularization: 1231.3856890201569%
written it to a csv file named tinyimagenetresnet18_500_0.1.csv.
```

K = 1000

```
Training Accuracy at training set on Client #999: 100.0%
Elapsing time for local training: 1120.0678660869598%
Aggregating!
Aggregation done!
Evaluating global model!
Elapsing time for training and aggregation: 1176.8097696304321%
Average accuracy on all Client: 54.71999740600586%
Average accuracy after regularization cleaning on all Client: 54.80999755859375%
Elapsing time plus cleansing regularization: 1198.2075035572052%
written it to a csv file named tinyimagenetresnet18_1000_0.1.csv.
```

https://colab.research.google.com/drive/1Hiqi46-bOY8U43E3KfudXAPEUm-u8aGp?usp=sharing

Reproduced Result

Invariance to Heterogeneity

Dateset	Setting	Accuracy
CIFAR - 10	Alpha = 0.1	82.5%
	Alpha = 0.05	82.5%
	S = 4	82.5%
	S = 2	82.5%
CIFAR - 100	Alpha = 0.1	58.50%
	Alpha = 0.05	58.50%
	S = 10	58.50%
	S = 5	58.50%
Tiny-ImageNet	Alpha = 0.1	54.80%
	Alpha = 0.05	54.80%
	S = 10	54.80%
	S = 5	54.80%

Invariance to Number of Clients

Dateset	No. of Clients	Accuracy
CIFAR - 100	K = 100	58.50%
	K = 500	58.50%
	K = 1000	58.50%
Tiny-ImageNet	K = 100	54.80%
	K = 500	54.80%
	K = 1000	54.80%

https://colab.research.google.com/drive/1Hiqi46-bOY8U43E3KfudXAPEUm-u8aGp?usp=sharing

Novelty Implemented

Gap in the Original AFL Paper

Key Insight Motivating Compression: In non-IID federated learning, each client sees only a small subset of classes. The local weight matrix w therefore has low intrinsic rank — most singular values are near zero. Transmitting the full matrix wastes bandwidth on near-zero components.

Problem	Detail
Full-rank W always transmitted	Even when clients hold very few classes (extreme non-IID), the full feat_size × num_classes matrix is sent
W grows with model scale	For ViT-B16 (feat_size = 768) with 200 classes, W per client = 0.234 MB × K clients
No compression mechanism	The original code has no way to trade off accuracy vs. communication cost
Silent rank-deficiency crashes	When Nk < feat_size (few samples per client), direct inversion fails with no fallback

Novelty Implemented

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Rank-r SVD Theory

- $W = U \cdot \Sigma \cdot V^T$
- where $U \in \mathbb{R}^{m \times m}$ is orthogonal (left singular vectors), $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal with non-negative singular values $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{min(m,n)} \geq 0$, and $V^T \in \mathbb{R}^{n \times n}$ is orthogonal (right singular vectors).
- The singular values tell you how much "energy" each component carries. If the matrix is truly rank-3, only $\sigma_1, \sigma_2, \sigma_3$ are non-zero. If it's approximately rank-3, $\sigma_1, \sigma_2, \sigma_3$ are large and $\sigma_4, \sigma_5, \ldots, \sigma_{100} \approx 0$.
- Singular Values = $\sqrt{}$(eigenvalues of W^TW)
- Big eigenvalue → big singular value → important direction → keep it.

Truncated (Rank-r) SVD keeps only the top-r components:

- $W \approx U_r \cdot diag(S_r) \cdot Vh_r$
- where $U_r \in \mathbb{R}^{512 \times r}$, $S_r \in \mathbb{R}^r$, $Vh_r \in \mathbb{R}^{r \times 100}$. This is provably the best rank-r approximation of W in the Frobenius norm sense the Eckart–Young theorem guarantees this. We are not making an arbitrary approximation; we are making the optimal one.
- We consider only Top r columns of U, Top r values of Σ and Top r rows of V^T.

Novelty Implemented

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Rank-r SVD Compression + Communication Cost Analyzer in AFL

Rank-r SVD Compression

- After each client computes its local weight matrix *w* (shape: *feat_size* × *num_classes*), truncated SVD is applied before aggregation.
- Only the top-r singular components are retained: $W \approx U_r \cdot diag(S_r) \cdot Vh_r$
- The matrix is reconstructed on the server side — the aggregation logic itself is unchanged
- Controlled by a single new argument: *--rank_r*

Communication Cost Analyzer

- Reports original vs. compressed cost, MB saved, and percentage savings

Novelty Implemented

CIFAR - 100

K = 100

```
Client 99 Train Acc: 94.43%
Local training time: 345.89s

Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)    : 512
  Num classes              : 100
  Num clients (K)          : 100
  Rank-r compression       : 32 (effective = 32)
  C matrix dominance       : 83.7% of total per-client cost
------------------------------------------------------------
                                  Original  Compressed
------------------------------------------------------------
  W matrix  (per client)          0.3906MB     0.1497MB
  C matrix  (per client)          2.0000MB     2.0000MB
  Total     (per client)          2.3906MB     2.1497MB
------------------------------------------------------------
  TOTAL     (all clients)       239.062MB    214.966MB
  W-only savings                   61.7%
  Overall savings                  10.1%  (24.097 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 367.63s
Global Accuracy: 58.53%

Cleaning regularization...
Accuracy after cleaning: 58.60%
Total time (with cleaning): 386.36s
Saved results to cifar100_resnet18_100_0.1.csv
```

K = 500

```
Client 499 Train Acc: 94.31%
Local training time: 536.41s

Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)    : 512
  Num classes              : 100
  Num clients (K)          : 500
  Rank-r compression       : 32 (effective = 32)
  C matrix dominance       : 83.7% of total per-client cost
------------------------------------------------------------
                                  Original  Compressed
------------------------------------------------------------
  W matrix  (per client)          0.3906MB     0.1497MB
  C matrix  (per client)          2.0000MB     2.0000MB
  Total     (per client)          2.3906MB     2.1497MB
------------------------------------------------------------
  TOTAL     (all clients)      1195.312MB   1074.829MB
  W-only savings                   61.7%
  Overall savings                  10.1%  (120.483 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 571.25s
Global Accuracy: 58.37%

Cleaning regularization...
Accuracy after cleaning: 58.50%
Total time (with cleaning): 591.70s
Saved results to cifar100_resnet18_500_0.1.csv
```

K = 1000

```
Client 999 Train Acc: 100.00%
Local training time: 754.36s

Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)    : 512
  Num classes              : 100
  Num clients (K)          : 1000
  Rank-r compression       : 32 (effective = 32)
  C matrix dominance       : 83.7% of total per-client cost
------------------------------------------------------------
                                  Original  Compressed
------------------------------------------------------------
  W matrix  (per client)          0.3906MB     0.1497MB
  C matrix  (per client)          2.0000MB     2.0000MB
  Total     (per client)          2.3906MB     2.1497MB
------------------------------------------------------------
  TOTAL     (all clients)      2390.625MB   2149.658MB
  W-only savings                   61.7%
  Overall savings                  10.1%  (240.967 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 798.11s
Global Accuracy: 58.21%

Cleaning regularization...
Accuracy after cleaning: 58.62%
Total time (with cleaning): 814.27s
Saved results to cifar100_resnet18_1000_0.1.csv
```

https://colab.research.google.com/drive/1-vVsTg4LklpF1cv70_MzQA7bD7Rw_RpW?usp=sharing

Novelty Implemented

Tiny-Imagenet

K = 100

```
Client 99 Train Acc: 81.98%
Local training time: 627.27s

Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)    : 512
  Num classes              : 200
  Num clients (K)          : 100
  Rank-r compression       : 32 (effective = 32)
  C matrix dominance       : 71.9% of total per-client cost
------------------------------------------------------------
                                  Original  Compressed
------------------------------------------------------------
  W matrix  (per client)          0.7812MB     0.1741MB
  C matrix  (per client)          2.0000MB     2.0000MB
  Total     (per client)          2.7812MB     2.1741MB
------------------------------------------------------------
  TOTAL     (all clients)       278.125MB    217.407MB
  W-only savings                   77.7%
  Overall savings                  21.8%  (60.718 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 650.83s
Global Accuracy: 54.97%

Cleaning regularization...
Accuracy after cleaning: 54.97%
Total time (with cleaning): 668.88s
Saved results to tinyimagenet_resnet18_100_0.1.csv
```

K = 500

```
Client 499 Train Acc: 84.19%
Local training time: 928.11s

Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)    : 512
  Num classes              : 200
  Num clients (K)          : 500
  Rank-r compression       : 32 (effective = 32)
  C matrix dominance       : 71.9% of total per-client cost
------------------------------------------------------------
                                  Original  Compressed
------------------------------------------------------------
  W matrix  (per client)          0.7812MB     0.1741MB
  C matrix  (per client)          2.0000MB     2.0000MB
  Total     (per client)          2.7812MB     2.1741MB
------------------------------------------------------------
  TOTAL     (all clients)      1390.625MB   1087.036MB
  W-only savings                   77.7%
  Overall savings                  21.8%  (303.589 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 966.48s
Global Accuracy: 54.84%

Cleaning regularization...
Accuracy after cleaning: 54.94%
Total time (with cleaning): 988.81s
Saved results to tinyimagenet_resnet18_500_0.1.csv
```

K = 1000

```
Client 999 Train Acc: 94.84%
Local training time: 1218.96s

Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)    : 512
  Num classes              : 200
  Num clients (K)          : 1000
  Rank-r compression       : 32 (effective = 32)
  C matrix dominance       : 71.9% of total per-client cost
------------------------------------------------------------
                                  Original  Compressed
------------------------------------------------------------
  W matrix  (per client)          0.7812MB     0.1741MB
  C matrix  (per client)          2.0000MB     2.0000MB
  Total     (per client)          2.7812MB     2.1741MB
------------------------------------------------------------
  TOTAL     (all clients)      2781.250MB   2174.072MB
  W-only savings                   77.7%
  Overall savings                  21.8%  (607.178 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 1269.52s
Global Accuracy: 54.94%

Cleaning regularization...
Accuracy after cleaning: 55.00%
Total time (with cleaning): 1288.84s
Saved results to tinyimagenet_resnet18_1000_0.1.csv
```

Novelty Implemented

Dateset	Setting	Accuracy	Communication cost saving on W	Percentage Saving on W
CIFAR - 100	K = 100	58.50%	24.097 MB	61.7%
	K = 500	58.50%	120.483 MB	61.7%
	K = 1000	58.50%	2149.658 MB	61.7%
Tiny Imagenet	K = 100	54.80%	60.718 MB	77.7%
	K = 500	54.80%	303.589 MB	77.7%
	K = 1000	54.80%	607.178 MB	77.7%

Conclusion and Future Work

My contribution extends AFL with:
Rank-r SVD compression on local weight matrices — reducing per-client upload cost by up to 87% + Communication cost analysis.

Three open directions identified from our analysis:

1. **Compress the C matrix —** The Gram matrix (*feat_size* × *feat_size*) dominates transmission cost (4.00 MB vs 0.39 MB for W in ResNet-18/CIFAR-100). C gets filled everywhere. No empty columns. No obvious sparsity. No clean low-rank structure.

2. **Partial participation and stragglers —**Adapting the AA law to handle asynchronous or dropout clients without sacrificing the invariance property is an open problem.

3. **Non-linear extension —** AFL is built on linear classifiers. Incorporating non-linear projections, kernel functions, or layer-wise least-squares formulations (label projection) could extend AFL to settings where the embedding space is not linearly separable.

References

1.Khodak et al. — *Federated Learning with Only Positive Labels* — uses low-rank structure in client updates. ICML 2022.

2.Zhao et al. — *FedPAGE: A Fast Local Stochastic Gradient Method for Communication-Efficient Federated Learning* — analyzes low-rank gradient compression. NeurIPS 2021.

3.Vogels et al. — *PowerSGD: Practical Low-Rank Gradient Compression for Distributed Optimization* — proposes rank-r approximation of gradient matrices via SVD for distributed training. NeurIPS 2019.

Thank You